

1. Product Search System

Code

```
import re

products = [
    "Apple iPhone 15",
    "Samsung Galaxy S24",
    "Apple MacBook Air",
    "Dell Laptop",
    "HP Pavilion",
    "Samsung Galaxy Book",
    "OnePlus Phone"
]

keyword = input("Enter search keyword: ")

# Exact search
exact = [
    p for p in products
    if re.search(r"\b" + re.escape(keyword) + r"\b", p, re.I)
]

# Prefix search
prefix = [
    p for p in products
    if re.search(r"\b" + re.escape(keyword), p, re.I)
]

# Suffix search
suffix = [
    p for p in products
    if re.search(re.escape(keyword) + r"\b", p, re.I)
]

# Partial search
partial = [
    p for p in products
    if re.search(re.escape(keyword), p, re.I)
]

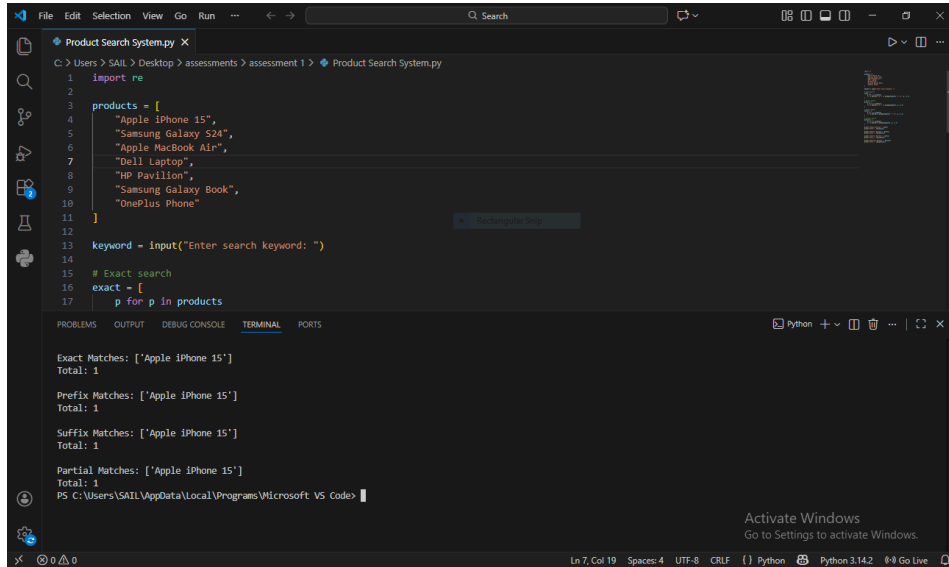
print("\nExact Matches:", exact)
print("Total:", len(exact))

print("\nPrefix Matches:", prefix)
print("Total:", len(prefix))

print("\nSuffix Matches:", suffix)
print("Total:", len(suffix))

print("\nPartial Matches:", partial)
print("Total:", len(partial))
```

Output



The image shows a screenshot of a Visual Studio Code (VS Code) editor window. The editor is open to a file named `Product Search System.py`. The code in the editor is as follows:

```
1 import re
2
3 products = [
4     "Apple iPhone 15",
5     "Samsung Galaxy S24",
6     "Apple MacBook Air",
7     "Dell Laptop",
8     "HP Pavilion",
9     "Samsung Galaxy Book",
10    "OnePlus Phone"
11 ]
12
13 keyword = input("Enter search keyword: ")
14
15 # Exact search
16 exact = [
17     p for p in products
```

The terminal window at the bottom shows the output of the program. It displays the results of a search for the keyword "Apple iPhone 15". The output is as follows:

```
Exact Matches: ['Apple iPhone 15']
Total: 1

Prefix Matches: ['Apple iPhone 15']
Total: 1

Suffix Matches: ['Apple iPhone 15']
Total: 1

Partial Matches: ['Apple iPhone 15']
Total: 1

PS C:\Users\SAIL\AppData\Local\Programs\Microsoft VS Code>
```

The status bar at the bottom of the VS Code window indicates the current line and column (Ln 7, Col 19), the number of spaces (4), the encoding (UTF-8), the line ending (CRLF), the language (Python), the Python version (Python 3.14.2), and the Go Live status.

2. Resume Information Extraction

Code

```
import re

resume = """
Name: Rahul Kumar
Email: rahul@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL, Machine Learning
Experience: 3 years
"""

# Extract name
name = re.search(
    r"Name:s*(.*)",
    resume
).group(1)

# Extract email
email = re.search(
    r"[\w.-]+@[ \w.-]+\.\w+",
    resume
).group()

# Extract mobile number
mobile = re.search(
    r"\b\d{10}\b",
    resume
).group()

# Extract skills
skills = [
    "Python",
    "Java",
    "SQL",
    "Machine Learning",
    "NLP"
]

found_skills = []

for skill in skills:
    if re.search(skill, resume, re.I):
        found_skills.append(skill)

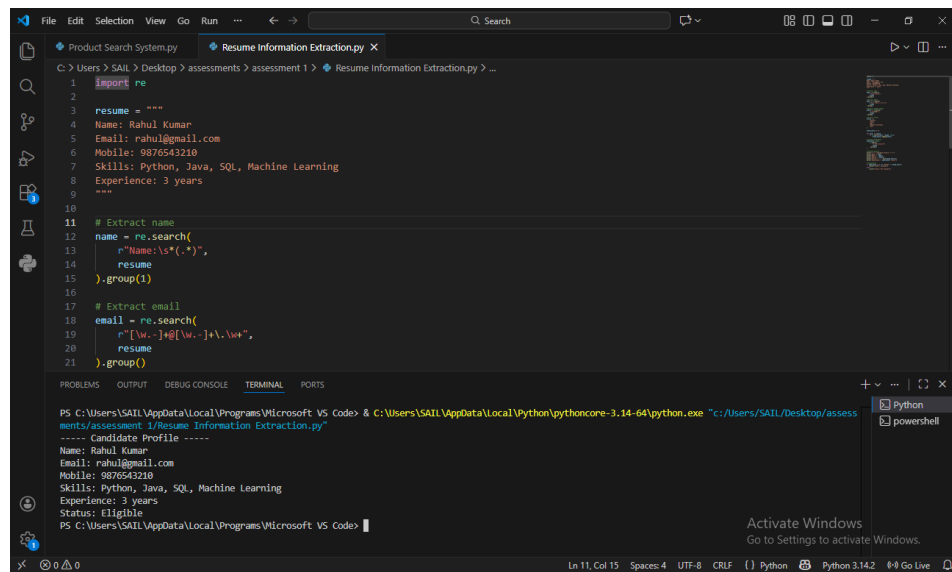
# Extract experience
experience = int(
    re.search(
        r"(\d+)\s*years?",
        resume
    ).group(1)
)

# Display profile
print("----- Candidate Profile -----")
print("Name:", name)
print("Email:", email)
print("Mobile:", mobile)
print("Skills:", ", ".join(found_skills))
print("Experience:", experience, "years")

# Eligibility
```

```
if experience >= 2 and "Python" in found_skills:
    print("Status: Eligible")
else:
    print("Status: Not Eligible")
```

Output



The screenshot shows a Visual Studio Code editor with a Python file named 'Resume Information Extraction.py'. The script defines a resume string and uses regular expressions to extract the name and email. The terminal output shows the extracted information and a status of 'Eligible'.

```
File Edit Selection View Go Run ... Search
Product Search System.py Resume Information Extraction.py X
C:\Users\SAIL\Desktop> assessments > assessment 1 > Resume Information Extraction.py ...
1 import re
2
3 resume = """
4 Name: Rahul Kumar
5 Email: rahul@gmail.com
6 Mobile: 9876543210
7 Skills: Python, Java, SQL, Machine Learning
8 Experience: 3 years
9 """
10
11 # Extract name
12 name = re.search(
13     r"Name:(.*)",
14     resume
15 ).group(1)
16
17 # Extract email
18 email = re.search(
19     r"[\w.-]+@([\w.-]+\.\w+)",
20     resume
21 ).group(1)
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```

```
PS C:\Users\SAIL\AppData\Local\Programs\Microsoft VS Code> & C:\Users\SAIL\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/SAIL/Desktop/assessments/assessment 1/Resume Information Extraction.py"
----- Candidate Profile -----
Name: Rahul Kumar
Email: rahul@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL, Machine Learning
Experience: 3 years
Status: Eligible
PS C:\Users\SAIL\AppData\Local\Programs\Microsoft VS Code>
```

Activate Windows
Go to Settings to activate Windows.

Ln 11, Col 15 Spaces: 4 UTF-8 CRLF Python Python 3.14.2 Go Live

3. University Registration System

Code

```
import re

reg = input("Enter Register Number: ")
email = input("Enter Email: ")
course = input("Enter Course Code: ")
semester = input("Enter Semester: ")
mobile = input("Enter Mobile Number: ")

# Validation
reg_valid = re.fullmatch(
    r"\d{2}[A-Z]{3}\d{3}",
    reg
)

email_valid = re.fullmatch(
    r"[\w.-]+@saveetha\.com",
    email
)

course_valid = re.fullmatch(
    r"[A-Z]{3}\d{3}",
    course
)

semester_valid = re.fullmatch(
    r"Semester [1-8]",
    semester,
    re.I
)

mobile_valid = re.fullmatch(
    r"[6-9]\d{9}",
    mobile
)

print("\n----- Validation Report -----")

print(
    "Register Number:",
    "Valid" if reg_valid else "Invalid"
)

print(
    "Email:",
    "Valid" if email_valid else "Invalid"
)

print(
    "Course Code:",
    "Valid" if course_valid else "Invalid"
)

print(
    "Semester:",
    "Valid" if semester_valid else "Invalid"
)

print(
    "Mobile Number:",
    "Valid" if mobile_valid else "Invalid"
```

)

Final status

if (

```
    reg_valid
    and email_valid
    and course_valid
    and semester_valid
    and mobile_valid
```

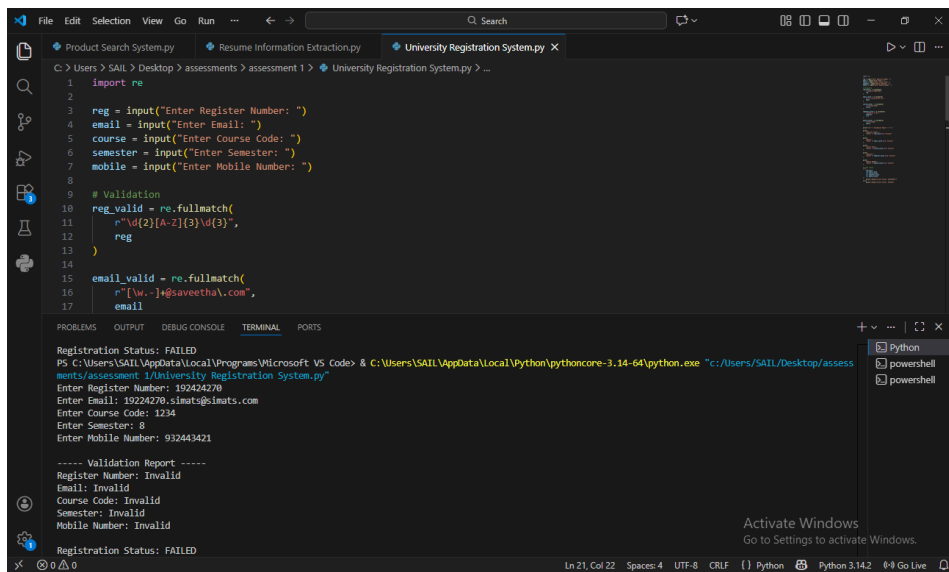
):

```
    print("\nRegistration Status: SUCCESSFUL")
```

else:

```
    print("\nRegistration Status: FAILED")
```

Output



```
File Edit Selection View Go Run ... Search
Product Search System.py Resume Information Extraction.py University Registration System.py X
C:\Users\SAIL\Desktop> assessments > assessment 1 > University Registration System.py > ...
1 import re
2
3 reg = input("Enter Register Number: ")
4 email = input("Enter Email: ")
5 course = input("Enter Course Code: ")
6 semester = input("Enter Semester: ")
7 mobile = input("Enter Mobile Number: ")
8
9 # Validation
10 reg_valid = re.fullmatch(
11     r"^\d{2}[A-Z]{3}\d{3}$",
12     reg
13 )
14
15 email_valid = re.fullmatch(
16     r"^[w.-]*@saveetha\.com",
17     email
18 )
19
20 course_valid = re.fullmatch(
21     r"^[A-Z]{3}\d{3}$",
22     course
23 )
24 semester_valid = re.fullmatch(
25     r"^[1-8]$",
26     semester
27 )
28 mobile_valid = re.fullmatch(
29     r"^\d{10}$",
30     mobile
31 )
32
33 # Final status
34 if (
35     reg_valid
36     and email_valid
37     and course_valid
38     and semester_valid
39     and mobile_valid
40 ):
41     print("\nRegistration Status: SUCCESSFUL")
42 else:
43     print("\nRegistration Status: FAILED")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Registration Status: FAILED
PS C:\Users\SAIL\AppData\Local\Programs\Microsoft VS Code> & C:\Users\SAIL\AppData\Local\Python\pythoncore-3.14-64\python.exe "C:\Users\SAIL\Desktop\assessments\assessment 1\University Registration System.py"
Enter Register Number: 192424270
Enter Email: 1924270.simat@simat.com
Enter Course Code: 1234
Enter Semester: 8
Enter Mobile Number: 932443421

----- Validation Report -----
Register Number: Invalid
Email: Invalid
Course Code: Invalid
Semester: Invalid
Mobile Number: Invalid

Registration Status: FAILED
```

Activate Windows
Go to Settings to activate Windows.

Ln 21, Col 22 Spaces: 4 UTF-8 CRLF Python Python 3.142 Go Live